INFINITESIMAL AUTOMORPHISMS BASE-CHANGE ALONG FLAT MAPS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1565-1572 of sources/stacks-stacks-more-morphisms/pinned-original-source.tex; proof span is lines 1574-1604, the complete author proof (29 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 34 of queue_stacks-stacks-more-morphisms.json (pb=1350, stmt 1565-1572, proof 1574-1604).

Editorial note. stacks-stacks-more-morphisms wave smm01 (2026-10-14): fresh adjudication accept per lane-s3/smm_judgments.md; pool per Chief 10:5x open-pool ruling (net 59 with-proof rows, bloodline lane-s3 by anchor audit, v2 true-key namespace stacks:stacks-more-morphisms.tex, row lemma-deform-property-fp-over-ft avoided on cross-chapter key collision); 60 with-proof rows judged 45 accept / 14 reject / 1 avoid; patch-sd-envs contract spans scanned.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 087R: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1604–1604. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 05ZC: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1604–1604. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 05ZF: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1604–1604. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0617: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1604–1604. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:stacks-more-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $\mathcal{X}$ be an algebraic stack over a scheme S. Assume $\mathcal{I}_\mathcal{X} \to \mathcal{X}$ is locally of finite presentation. Let $A \to B$ be a flat S-algebra homomorphism. Let x be an object of $\mathcal{X}$ over A and set $y = x|_B$. Then $Inf_x(M) \otimes_A B = Inf_y(M \otimes_A B)$.*

Proof. Recall that $\mathrm{Inf}_x(M)$ is the set of automorphisms of the trivial deformation of x to $A[M]$ which induce the identity automorphism of x over A. The trivial deformation is the pullback of x to $\mathrm{Spec}(A[M])$ via $\mathrm{Spec}(A[M]) \to \mathrm{Spec}(A)$. Let $G \to \mathrm{Spec}(A)$ be the automorphism group algebraic space of x (this exists because $\mathcal{X}$ is an algebraic space). Let $e : \mathrm{Spec}(A) \to G$ be the neutral element. The discussion in More on Morphisms of Spaces, Section 0617 gives

$$\mathrm{Inf}_x(M) = \mathrm{Hom}_A(e^*\Omega_{G/A}, M)$$

By the same token

$$\mathrm{Inf}_y(M \otimes_A B) = \mathrm{Hom}_B(e_B^*\Omega_{G_B/B}, M \otimes_A B)$$

Since $G \to \mathrm{Spec}(A)$ is locally of finite presentation by assumption, we see that $\Omega_{G/A}$ is locally of finite presentation, see More on Morphisms of Spaces, Lemma 05ZF. Hence $e^*\Omega_{G/A}$ is a finitely presented A-module. Moreover, $\Omega_{G_B/B}$ is the pullback of $\Omega_{G/A}$ by More on Morphisms of Spaces, Lemma 05ZC. Therefore $e_B^*\Omega_{G_B/B} = e^*\Omega_{G/A} \otimes_A B$. we conclude by More on Algebra, Lemma 087R. $\square$